

Research Unplugged: Introduction to R & RStudio

Sam J. Vizcaino Vickers
Psychology Technician
Department of Psychology, Wrexham University

✉ `sam.vickers@wrexham.ac.uk`
✉ `psychology.technician@wrexham.ac.uk`
✉

What is R?

- ▶ Open-source language designed for **statistical computing, analysis and open science**.
- ▶ Extensive Package Ecosystem: features over 20,000 packages (via CRAN) that allow users to perform several different statistical models and visualisations.
- ▶ Renowned for its ability to produce high quality graphics and interactive web apps

What is RStudio?

- ▶ It is an integrated development environment (IDE)
- ▶ Built for Data Science
- ▶ Workspace for Scripting, academic Manuscripts, Presentations, and more...

Alternatives?

- ▶ What about alternative programs such Python, SPSS, SAS, Stata, Minitab, Matlab, Maple, Mathematica, Tableau, Excel, SQL and many others) come down to 3 inter-related factors:
 - ▶ It is immensely powerful
 - ▶ Open source
 - ▶ Very and increasingly widely used
- ▶ Positron (IDE)
 - ▶ New upcoming IDE
 - ▶ Worth it?

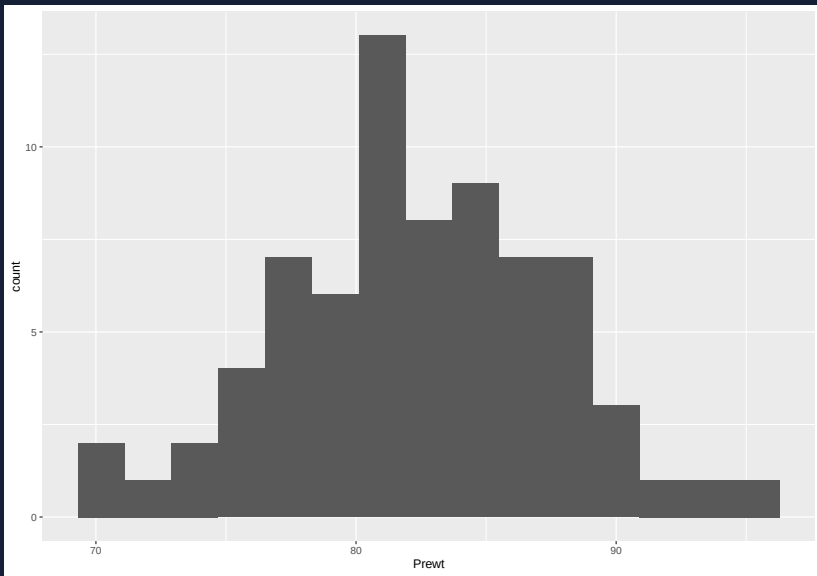
Essential Concepts

- ▶ Data Wrangling
- ▶ Data Visualisation
- ▶ Quarto
- ▶ Statistics

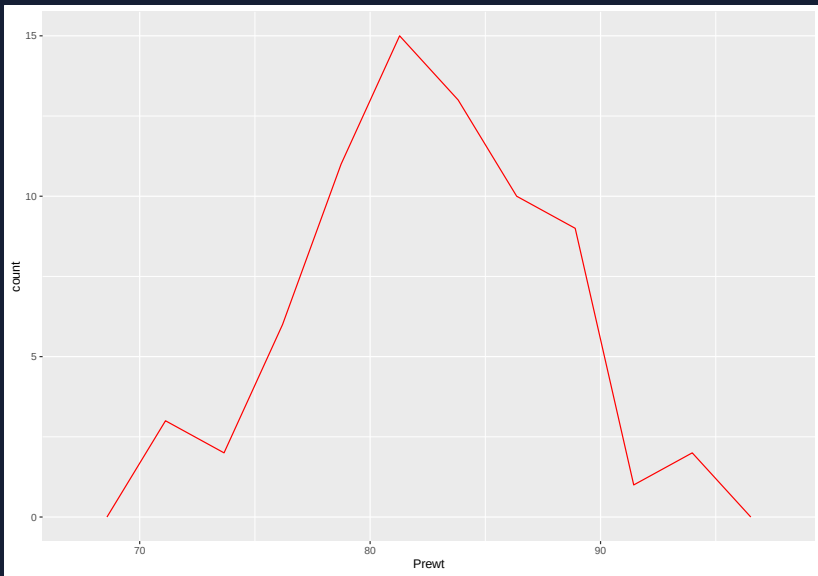
What are we going to learn today

- ▶ *Guided tour of RStudio*
 - ▶ Calling R packages, functions, variables etc.
- ▶ *First Steps in R*
- ▶ Basic Data visualisations

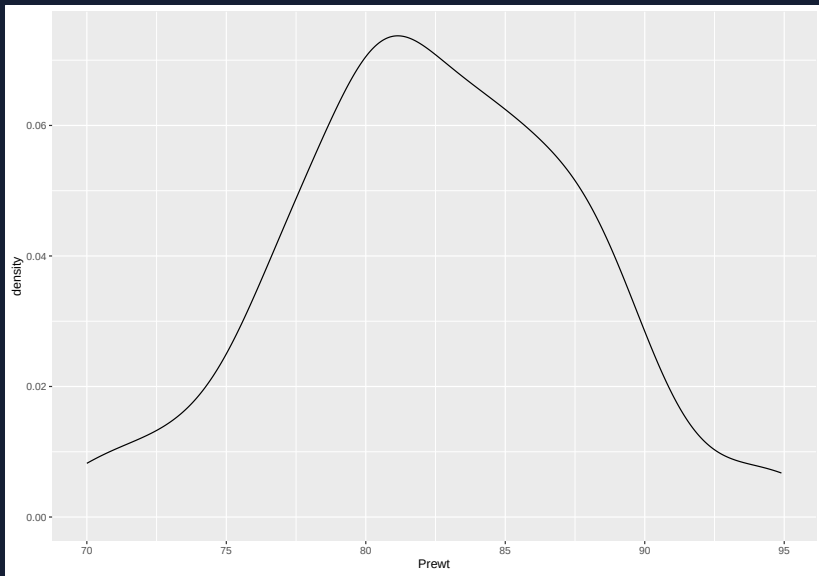
Histograms



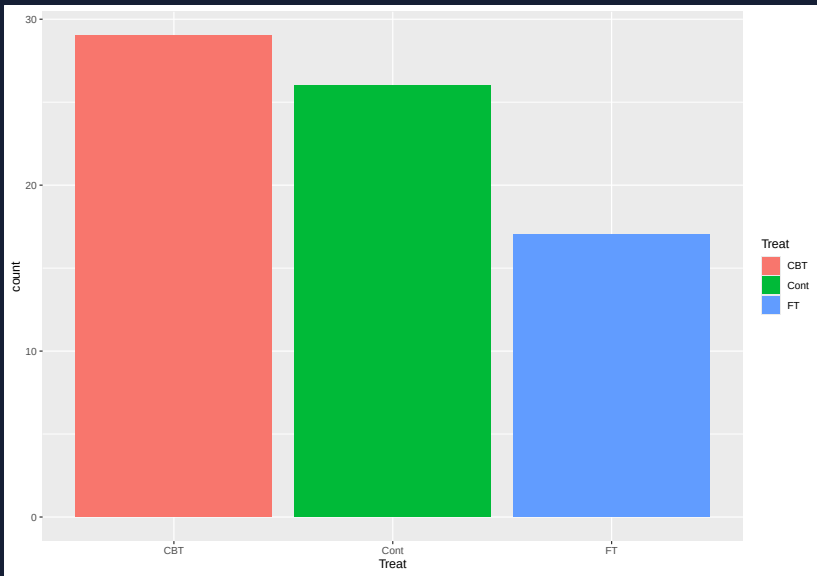
Frequency Polygons



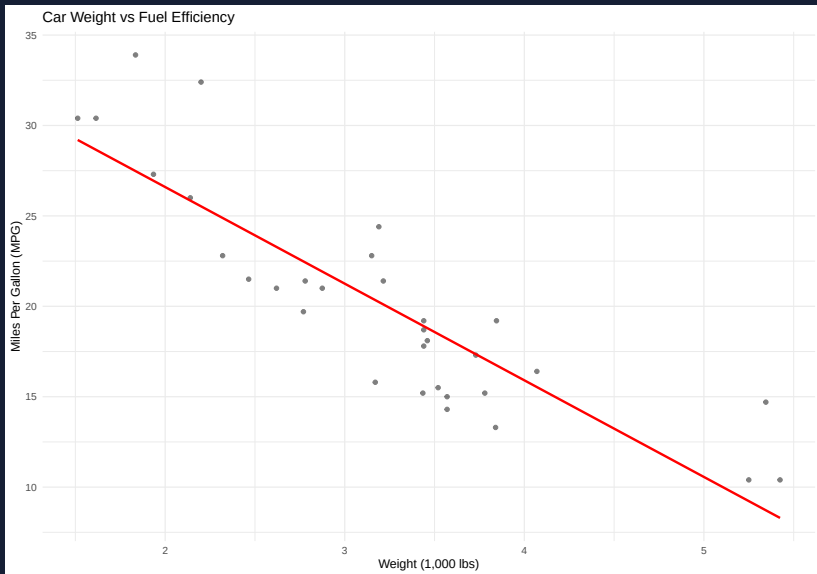
Density Plots



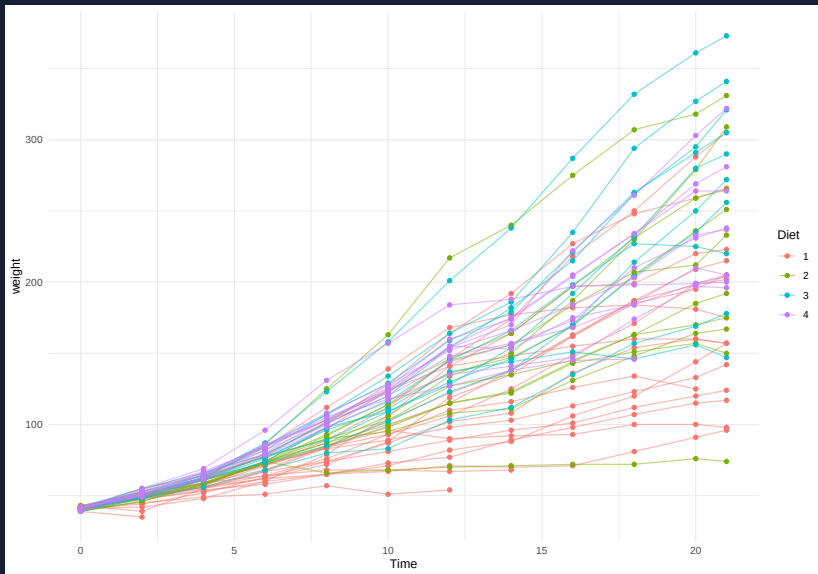
Bar Plots



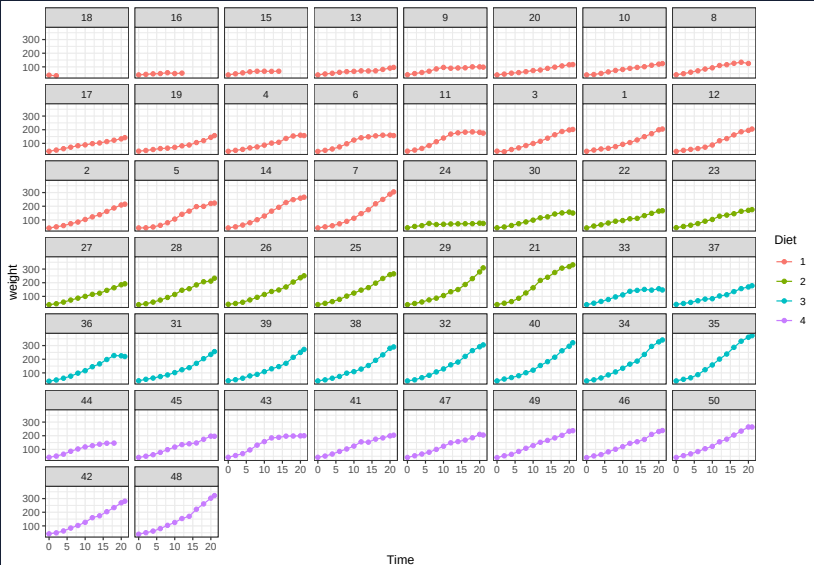
Scatter Plot



Growth Curve Plots



Facet Plots



Psychology Students

For any psychology Students who wish to learn more about R and RStudio then do book a session with us on the Booking System.